



LM Farms Dairy Platform Spring 2023 Autumn 2024



Customer Name: L M Farms
Address: 553 West Coast Road
RD 1
Dargaville

Phone:
Mobile: 021 723-940
Fax:
E-mail: lmfarms@xtra.co.nz

Farm Type: Dairy
Crop: ☒ Pasture
Soil Type(s): ☒ Sedimentary
Area (ha Tot): 186

Version Number: 5.0
Representative: Russell Plank
Telephone: 021 752-653
Date: 3-Mar-26

Date	Area (ha eff)	Spread Rate (Kg/Ha)	Fertiliser Product	Percent of Mix	Trace Elements	Kg/T	Nutrients Applied Kg/Ha						Tonnes Req	\$/Tonne product	\$/Tonne cart & spread	Total \$	Balance Order Number			
							N	P	K	S	Mg	Ca								
Spring 2023 Dairy Platform Early Spring 2023	170	80	N-rich ammo 36N	100%			29	0	0	8	0	0	13.6	\$774		\$10,520				
							Despatch ex Ballance Dargaville													
Spring 2023 Dairy Platform Non Effluent Block	120	100	Sustain 20K	100%			28	0	20	0	0	0	12.0	\$1,111		\$13,329				
							Despatch ex Ballance Dargaville													
Spring 2024 Dairy Platform Effluent	50	60	Sustain Green	100%			28	0	0	17	0	0	3.0	\$859		\$2,578				
							Despatch ex Ballance Dargaville													
Autumn 2024 Dairy Platform	170	100	PhaSed N QS	100%			31	0	0	18	0	2	17.0	\$632		\$10,737				
							Despatch ex Ballance Dagaville													
YTD Nutrients Applied Over:							80	0	13	28	0	2	46					\$37,164		
Maintenance Requirements for							1600	Kg	MS/ha	-	81	104	96	31	36					YTD \$200 per Ha

Prices based on delivery: **FALSE** unless otherwise stated.

COMMENTS:

All pricing is based on current day prices, exclusive of GST as of the date stated above. Fertiliser pricing is subject to change and will be invoiced based on prices ruling on day of delivery

Page 1

Our technical advice is given in good faith but without warranty. The application and use of products is beyond our control and we therefore do not warrant pasture, plant or crop performance to any specific level. In making recommendations we rely on information provided to us by the customer and/or third parties such as commercial analytical services. As an animal health precaution, do not allow livestock to graze pasture until fertiliser has disappeared from foliage. Mixes containing nitrogen can become unstable and should be applied either separately or immediately upon delivery. Please refer to the Fertiliser Codes of Practice with respect to both third party and environmental risk effects.



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Getting the Best out of your soil

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Farm Type: Dairy
Crop: Pasture
Soil Type(s): Sedimentary
Area (ha Tot): 186

Version Number: 5.0
Representative: Russell Plank
Telephone: 021 752-653
Date: 9-Feb-26

Date	Area (ha eff)	Spread Rate (Kg/Ha)	Fertiliser Product	Percent of Mix	Trace Elements	Kg/T	Nutrients Applied Kg/Ha						Tonnes Req	\$/Tonne product	\$/Tonne cart & spread	Total \$	Balance Order Number
							N	P	K	S	Mg	Ca					
Spring 2024 Dairy Platform Non Effluent	101	560	Serpentine Super Muriate of Potash Sustain Green	80% 10% 10%			25	32	28	36	23	77	56.6	\$766		\$43,356	
							Despatch ex Balance Dargaville										
Spring 2024 Dairy Platform Old Effluent	60	350	Sustain Green Serpentine Super	16% 84%			25	21	0	23	15	50	21.0	\$572		\$12,012	
							Despatch ex Balance Dargaville										
Spring 2024 Dairy Platform New Effluent	25	400	Sustain Green Muriate of Potash Serpentine Super	14% 10% 76%			25	22	20	24	15	52	10.0	\$601		\$6,010	
							Despatch ex Balance Dargaville										
Autumn 2025 Dairy Platform	130	110	PhaSed N #	100%			28	0	0	31	0	0	14.3	\$1,175		\$16,803	
YTD Nutrients Applied Over:							45	27	18	52	19	65	102			\$78,181	
Maintenance Requirements for							-	81	104	96	31	36				YTD \$420	per Ha

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E-mail: lmfarms@xtra.co.nz

Farm Type:

Dairy

Crop:

Pasture

Soil Type(s):

Sedimentary

Area (ha Tot): 186

Version Number 5.0

Representative: Russell Plank

Telephone 021 752-653

Date: 3-Mar-26

Date	Area (ha eff)	Spread Rate (Kg/Ha)	Fertiliser Product	Percent of Mix	Trace Elements	Kg/T	Nutrients Applied Kg/Ha						Tonnes Req	\$ /Tonne product	\$ /Tonne cart & spread	Total \$	Balance Order Number
							N	P	K	S	Mg	Ca					
Non Effluent LM Farms Platform Spring	101	560	Serpentine Super Muriate of Potash Sustain Green	80% 10% 10%			25	31	28	38	23	77	56.6	\$667		\$37,725	
Old Effluent LM Farms Dairy Platform Spring	60	350	Serpentine Super Sustain Green	84% 16%			25	20	0	25	15	50	21.0	\$648		\$13,616	
New Effluent LM Farms Platform Spring	25	390	Serpentine Super Muriate of Potash Sustain Green	76% 10% 14%			25	20	20	25	15	50	9.8	\$686		\$6,722	
YTD Nutrients Applied Over:							25	26	18	32	19	65	87			\$58,063	
Maintenance Requirements for							-	81	104	96	31	36				YTD \$312	per Ha

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Please refer to the Fertiliser Codes of Practice with respect to both third party and environmental risk effects.

Certificate of Analysis

Page 1 of 6

Client:	LM Farms	Lab No:	3696148	shvmpv1
Address:	C/- Michael Slagter 9 Smith-Grey Crescent RD 4 Hikurangi 0184	Date Received:	17-Oct-2024	
		Date Reported:	24-Oct-2024	
		Quote No:		
		Order No:		
		Client Reference:	Dairy Platform S/T 2024	
		Submitted By:	Russell Plank	

Sample Name: Non Effluent (12 13 34 35)

Lab Number: 3696148.1

Sample Type: SOIL Mixed Pasture, Dairy (Sed.) (S181)

Analysis	Level Found	Medium Range*	Low	Medium	High
pH	pH Units	5.6	5.8 - 6.2		
Olsen Phosphorus	mg/L	43	20 - 30		
Anion Storage Capacity*	%	9			
Potassium	MAF units	3	6 - 8		
Calcium	MAF units	8	4 - 10		
Magnesium	MAF units	26	8 - 10		
Sodium	MAF units	5			
Sulphate Sulphur	mg/kg	5	10 - 12		
Extractable Organic Sulphur*	mg/kg	7	15 - 20		
Soil Sample Depth**	mm	0-75			
Soil Type**		Sedimentary			
Base Saturation %		K 1.0 Ca 47 Mg 8.2 Na 0.7			
me/100g		K 0.17 Ca 7.6 Mg 1.31 Na 0.11			
Additional Properties		Cation Exchange Capacity (me/100g) 16			
		Total Base Saturation (%) 57			
		Volume Weight (g/mL) 0.90			



This Laboratory is accredited by International Accreditation New Zealand (IANZ), which represents New Zealand in the International Laboratory Accreditation Cooperation (ILAC). Through the ILAC Mutual Recognition Arrangement (ILAC-MRA) this accreditation is internationally recognised. The tests reported herein have been performed in accordance with the terms of accreditation, with the exception of tests marked * or any comments and interpretations, which are not accredited.

Certificate of Analysis

Page 2 of 6

Client:	LM Farms	Lab No:	3696148	shvmpv1
Address:	C/- Michael Slagter 9 Smith-Grey Crescent RD 4 Hikurangi 0184	Date Received:	17-Oct-2024	
		Date Reported:	24-Oct-2024	
		Quote No:		
		Order No:		
		Client Reference:	Dairy Platform S/T 2024	
		Submitted By:	Russell Plank	

Sample Name: Old Effluent (41 42 43 51)

Lab Number: 3696148.2

Sample Type: SOIL Mixed Pasture, Dairy (Sed.) (S181)

Analysis	Level Found	Medium Range*	Low	Medium	High
pH	pH Units	6.0	5.8 - 6.2		
Olsen Phosphorus	mg/L	30	20 - 30		
Anion Storage Capacity*	%	14			
Potassium	MAF units	10	6 - 8		
Calcium	MAF units	7	4 - 10		
Magnesium	MAF units	48	8 - 10		
Sodium	MAF units	7			
Sulphate Sulphur	mg/kg	8	10 - 12		
Extractable Organic Sulphur*	mg/kg	5	15 - 20		
Soil Sample Depth**	mm	0-75			
Soil Type**		Sedimentary			
Base Saturation %		K 4.0 Ca 46 Mg 16.9 Na 1.2			
me/100g		K 0.56 Ca 6.5 Mg 2.37 Na 0.17			
Additional Properties		Cation Exchange Capacity (me/100g)	14		
		Total Base Saturation (%)	68		
		Volume Weight (g/mL)	0.91		

Certificate of Analysis

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Client:	LM Farms	Lab No:	3696148	shvmpv1
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		Date Reported:	24-Oct-2024	
		Quote No:		
		Order No:		
		Client Reference:	Dairy Platform S/T 2024	
		Submitted By:	Russell Plank	

Sample Name: New Effluent (34 35 59 60)

Lab Number: 3696148.3

Sample Type: SOIL Mixed Pasture, Dairy (Sed.) (S181)

Analysis	Level Found	Medium Range*	Low	Medium	High
pH	pH Units	6.2	5.8 - 6.2		
Olsen Phosphorus	mg/L	22	20 - 30		
Anion Storage Capacity*	%	< 3			
Potassium	MAF units	6	6 - 8		
Calcium	MAF units	8	4 - 10		
Magnesium	MAF units	30	8 - 10		
Sodium	MAF units	5			
Sulphate Sulphur	mg/kg	6	10 - 12		
Extractable Organic Sulphur*	mg/kg	6	15 - 20		
Soil Sample Depth**	mm	0-75			
Soil Type**		Sedimentary			
Base Saturation %		K 2.6 Ca 60 Mg 11.9 Na 1.0			
me/100g		K 0.34 Ca 7.8 Mg 1.55 Na 0.13			
Additional Properties		Cation Exchange Capacity (me/100g)	13		
		Total Base Saturation (%)	75		
		Volume Weight (g/mL)	0.87		

The above nutrient graphs compare the levels found with reference interpretation levels. NOTE: It is important that the correct sample type be assigned, and that the recommended sampling procedure has been followed. R J Hill Laboratories Limited does not accept any responsibility for the resulting use of this information. IANZ Accreditation does not apply to comments and interpretations, i.e. the 'Range Levels' and subsequent graphs.

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Page 4 of 6

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		Order No:		
		Client Reference:	Dairy Platform S/T 2024	
		Submitted By:	Russell Plank	

Soil Analysis Results

Sample Name:		Non Effluent (12 13 34 35)	Old Effluent (41 42 43 51)	New Effluent (34 35 59 60)			
Lab Number:		3696148.1	3696148.2	3696148.3			
Sample Type:		SOIL Mixed Pasture, Dairy (Sed.)	SOIL Mixed Pasture, Dairy (Sed.)	SOIL Mixed Pasture, Dairy (Sed.)			
Sample Type Code:		S181	S181	S181			
pH	pH Units	5.6	6.0	6.2	-	-	-
Olsen Phosphorus	mg/L	43	30	22	-	-	-
Anion Storage Capacity*	%	9	14	< 3	-	-	-
Potassium	me/100g	0.17	0.56	0.34	-	-	-
Potassium	%BS	1.0	4.0	2.6	-	-	-
Potassium	MAF units	3	10	6	-	-	-
Calcium	me/100g	7.6	6.5	7.8	-	-	-
Calcium	%BS	47	46	60	-	-	-
Calcium	MAF units	8	7	8	-	-	-
Magnesium	me/100g	1.31	2.37	1.55	-	-	-
Magnesium	%BS	8.2	16.9	11.9	-	-	-
Magnesium	MAF units	26	48	30	-	-	-
Sodium	me/100g	0.11	0.17	0.13	-	-	-
Sodium	%BS	0.7	1.2	1.0	-	-	-
Sodium	MAF units	5	7	5	-	-	-
CEC	me/100g	16	14	13	-	-	-
Total Base Saturation	%	57	68	75	-	-	-
Volume Weight	g/mL	0.90	0.91	0.87	-	-	-
Sulphate Sulphur	mg/kg	5	8	6	-	-	-
Extractable Organic Sulphur*	mg/kg	7	5	6	-	-	-
Soil Sample Depth**	mm	0-75	0-75	0-75	-	-	-
Soil Type**		Sedimentary	Sedimentary	Sedimentary	-	-	-

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		Submitted By:	Russell Plank	

Analyst's Comments

† Customer supplied data. Please note: Hill Labs cannot be held responsible for the validity of this customer supplied data, or any subsequent calculations that rely on this information.

Samples 1-3 Comment:

The medium or optimum range guidelines shown in the histogram report relate to sampling protocols as per Hill Laboratories' crop guides and are based on reference values where these are published. Results for samples collected to different depths than those described in the crop guide should be interpreted with caution. For pastoral soils, the medium ranges are specific for a 75mm sample depth, but if a 150mm sampling depth is used the nutrient levels measured may appear low against these ranges, as nutrients are typically more concentrated in the top of the soil profile. These soil profile differences are altered upon cultivation or contouring. Further explanation of the derivation of the medium and optimum ranges is available on request.

Samples 1-3 Comment:

While soil Mg MAF levels of 8-10 (0.4 - 0.6 me/100g) are sufficient for pasture production, soil levels of 25-30 (1 - 1.6 me/100g) are required to ensure adequate Mg content in pasture for animal health (greater than 0.22% in the herbage).

Samples 1-3 Comment:

Anion Storage Capacity (also known as Phosphate Retention) is an inherent property of the soil type and does not change. Phosphorus and sulphur fertiliser recommendations should take this value into account. Soils may be classified as Low (less than 30%), Medium (30-60%) or High (greater than 60%) ASC.

Samples 1-3 Comment:

For intensive farm systems with high stocking-rate and/or high-production/ha, increasing the soil Olsen P optimum ranges to 30-40 (ash and sedimentary soils) and 45-55 (pumice and peat soils) may be justified.

Summary of Methods

The following table(s) gives a brief description of the methods used to conduct the analyses for this job. The detection limits given below are those attainable in a relatively simple matrix. Detection limits may be higher for individual samples should insufficient sample be available, or if the matrix requires that dilutions be performed during analysis. A detection limit range indicates the lowest and highest detection limits in the associated suite of analytes. A full listing of compounds and detection limits are available from the laboratory upon request. Unless otherwise indicated, analyses were performed at Hill Labs, 28 Duke Street, Frankton, Hamilton 3204.

Sample Type: Soil			
Test	Method Description	Default Detection Limit	Sample No
Sample Registration*	Samples were registered according to instructions received.	-	1-3
Soil Prep (Dry & Grind)*	Air dried at 35 - 40°C overnight (residual moisture typically 4%) and crushed to pass through a 2mm screen.	-	1-3
pH	1:2 (v/v) soil:water slurry followed by potentiometric determination of pH. In-house.	0.1 pH Units	1-3
Olsen Phosphorus	Olsen extraction followed by Molybdenum Blue colorimetry. In-house method.	1 mg/L	1-3
Sulphate Sulphur	0.02M Potassium phosphate extraction followed by Ion Chromatography. In-house.	1 mg/kg	1-3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 MAF units	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	2 MAF units	1-3
Extractable Organic Sulphur*	Determined by NIR, calibration based on; 0.02M Potassium phosphate extraction. Total extractable S determined by ICP-OES from which the Sulphate-S is subtracted.	2 mg/kg	1-3

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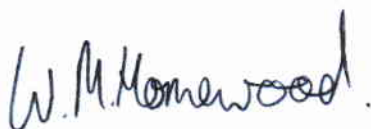
Sample Type: Soil			
Test	Method Description	Default Detection Limit	Sample No
Anion Storage Capacity*	Determined by NIR, calibration based on; Equilibration with 1000 mg/L P solution followed by colorimetric analysis.	10 %	2
Anion Storage Capacity	Equilibration with 1000 mg/L P solution followed by colorimetric analysis. In-house.	3 %	1, 3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.01 me/100g	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.5 me/100g	1-3
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.04 me/100g	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.05 me/100g	1-3
Potassium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1-3
Calcium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	1 %BS	1-3
Magnesium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.2 %BS	1-3
Sodium	1M Neutral ammonium acetate extraction followed by ICP-OES. In-house.	0.1 %BS	1-3
CEC	Summation of extractable cations (K, Ca, Mg, Na) and extractable acidity. May be overestimated if soil contains high levels of soluble salts or carbonates. In-house.	2 me/100g	1-3
Total Base Saturation	Calculated from Extractable Cations and Cation Exchange Capacity.	5 %	1-3
Volume Weight	The weight/volume ratio of dried, ground soil. In-house.	0.01 g/mL	1-3

These samples were collected by yourselves (or your agent) and analysed as received at the laboratory.

Testing was completed between 17-Oct-2024 and 23-Oct-2024. For completion dates of individual analyses please contact the laboratory.

Samples are held at the laboratory after reporting for a length of time based on the stability of the samples and analytes being tested (considering any preservation used), and the storage space available. Once the storage period is completed, the samples are discarded unless otherwise agreed with the customer. Extended storage times may incur additional charges.

This certificate of analysis must not be reproduced, except in full, without the written consent of the signatory.



Wendy Homewood
Operations Support - Agriculture